

# Understanding and Predicting Bitcoin Crashes with Probit Models

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## Abstract

Predicting cryptocurrency crashes is challenging due to their speculative dynamics, extreme volatility, and limited regulatory structure. This study investigates the predictability of Bitcoin crashes using standard and dynamic probit models applied to daily data from 2017 to 2023. Crash indicators are constructed using a  $3.09\sigma$  tail event rule, and model performance is evaluated through an extensive grid search over multiple crash horizons and consolidation windows, ensuring robustness against horizon dependent distortions. The empirical results show that hybrid models that combine sentiment, macro-financial variables, and Bitcoin specific returns consistently outperform sentiment only specifications.

The findings highlight the importance of combining behavioral indicators with global risk measures and cryptocurrency specific dynamics to capture the multifaceted drivers of Bitcoin crashes. The results have implications for investors, exchanges, and policymakers looking for early

warning mechanisms for systemic risk in digital asset markets.

**Keywords:** Bitcoin, crashes, probit models, cryptocurrency

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## 1. Introduction

Predicting a crash in financial markets is important since such events can have lasting consequences for investors and the broader economy. Financial crashes often lead to significant losses in wealth, increased unemployment, economic recessions, and political implications (Reinhart et al. 2009), as was evident in the 2008 global financial crisis. This crash, triggered by the housing bubble's collapse and the failure of major financial institutions, led to trillions of dollars in lost wealth, widespread job losses, and a severe economic downturn that affected economies worldwide. In cryptocurrency markets, predicting crashes becomes even more critical due to their highly volatile and speculative nature. Cryptocurrencies operate with few regulatory safeguards, which exposes the market to large, unanticipated price fluctuations, and market manipulation. For example, the cryptocurrency market crash in early 2018, where Bitcoin lost nearly 65% of its value within a month, wiped out billions of dollars in investments and caused widespread panic among investors (Manahov 2024). Predicting such crashes in the cryptocurrency market is very important for protecting individual investors from major losses, and maintaining the stability of the financial system as cryptocurrencies become increasingly integrated into mainstream finance.

This paper forecasts future Bitcoin crashes by using diverse indicators including market sentiment, economic signals, and blockchain activity. This research addresses a core question: How effective are different predictive models in forecasting Bitcoin crashes considering various market indicators?

The contributions of this study are threefold. First, a novel robustness based crash identification framework that systematically compares horizons, thresholds, and consolidation windows, providing a standardized method for defining cryptocurrency crashes. We propose a crash-identification framework based on a  $3.09\sigma$  tail-risk rule combined with a systematic grid search over multiple crash horizons and consolidation windows. This approach addresses inconsistencies in definitions of cryptocurrency crashes and ensures economically meaningful and statistically robust event labeling.

Second, we provide a comparative assessment of behavioral, macro-financial, and cryptocurrency specific predictors, showing that hybrid models, particularly those combining Google search

intensity, S&P 500, VIX, and Bitcoin returns, substantially outperform sentiment only specifications.

Third, we demonstrate that crash predictability in cryptocurrency markets is highly horizon dependent, meaningful discriminatory power emerges only at intermediate horizons of 10 to 30 days, while very short horizons exhibit no predictive content. This highlights the importance of horizon selection when evaluating crash risk in high volatility digital asset markets.

The use of the probit and dynamic probit models is well motivated by the binary nature of the crash variable. The outcome  $y_t$  indicates whether a crash occurs within a specified horizon, making the probit model an appropriate choice for mapping predictors into event probabilities. Probit models have a long established track record in forecasting recession risks, financial stress, and tail events (Chin et al. 2020; Christiansen et al. 2014; Hausman et al. 1992; W et al. 2012). By applying this methodology to Bitcoin, we extend a proven econometric framework to a market where predictive modelling remains underdeveloped. Moreover, the dynamic probit enhances predictive performance by incorporating lagged latent states and allowing key variables to exhibit temporal dependence, thereby aligning the econometric structure with cryptocurrency market dynamics.

Market sentiment remains an important dimension in our analysis, yet our results demonstrate that sentiment alone is insufficient for strong crash prediction. While Google search intensity contributes meaningfully, the best performing models are those that combine sentiment with global risk indicators and Bitcoin-specific returns. This finding contrasts with studies that emphasize sentiment as the primary driver of cryptocurrency movements (Soloviev et al. 2018). Instead, our evidence suggests that cryptocurrency crashes are jointly driven by behavioral signals, cross-market spillovers, and volatility dynamics, highlighting the need for an integrated modeling approach. By comparing the predictors used in traditional stock market crash studies such as volume trends, return skewness, and investor attention (Pyrlik 2013; Chen et al. 2001; Xu et al. 2021; Zhou et al. 2003) with those in cryptocurrency markets, we show that the latter rely more heavily on behavioral variables and global risk indicators.

Our study uses daily data from January 1, 2017 till December 31, 2023, 2556 observation, a

period that captures major crash episodes, as well as significant structural changes and regulatory developments in cryptocurrency markets. This sample allows us to evaluate predictive performance in an environment characterized by high volatility and substantial market evolution. The dataset integrates predictors from Google trends, S&P500 Index (SPX), Volatility Index (VIX), and Bitcoin (BTC) returns, as well as factor variables capturing financial and economic indicators, enabling us to examine both behavioral and macro-financial determinants of crashes.

Overall, the insights of this paper contribute to the understanding of crash predictability in digital-asset markets and offer practical implications for investors, regulators, and researchers. The remainder of the paper is structured as follows: Section 2 reviews the related literature; Section 3 presents the methodology and dataset; Section 4 reports the results; and Section 5 concludes.

## **2. Literature Review**

### *2.1. Crash Prediction in Cryptocurrency Markets*

Crash prediction has long been examined in traditional financial markets, where researchers identify a range of behavioral, volatility based, and macro-financial indicators that precede extreme downside events. Early work by Chen et al. (2001) and Zhou et al. (2003) shows that abnormal trading volume, negative skewness, and speculative pressures significantly increase the likelihood of stock price crashes. More recently, sentiment and attention based indicators have become central in forecasting extreme returns. Xu et al. (2021) demonstrate that Google search intensity provides leading information about impending equity market crashes, while Guan et al. (2022) show that real-time social media sentiment improves crash prediction during periods of acute market stress such as the COVID-19.

Binary-response forecasting models, especially probit and dynamic probit approaches, have been widely used in recession and financial crisis prediction (Nyberg 2010; Kauppi et al. 2008; U et al. 2020; Antunes et al. 2018). These studies highlight the importance of latent state dependence, time-varying volatility, and macro-financial drivers in anticipating extreme events. Their insights are directly relevant for cryptocurrency markets, which exhibit substantial regime shifts, clustering

of volatility, and sensitivity to global risk conditions.

In the cryptocurrency domain, recent work is increasingly focusing on the joint roles of sentiment, volatility spillovers, and macro–financial linkages. W et al. (2021) and A et al. (2021) find that investor attention and sentiment indices improve the forecasting of cryptocurrency volatility and downside tail events, although sentiment alone does not fully explain crash formation. At the same time, volatility spillovers play a crucial role. Elie et al. (2019) show that major cryptocurrencies exhibit strong volatility connectedness, while Kang et al. (2022) document that spillovers are asymmetric, with stronger transmission during crash periods. These findings highlight that crash risk is not isolated to individual assets but spreads throughout the digital asset ecosystem.

Macro–financial interactions have also emerged as key determinants of cryptocurrency crashes. Yousaf et al. (2022) show that global equity market shocks transmit strongly to Bitcoin, particularly during stress episodes, while Conlon et al. (2020) and Shaen et al. (2020) demonstrate that cryptocurrencies experience severe drawdowns during global uncertainty events, such as the COVID19 pandemic. The accumulation of this evidence strongly suggests that models excluding macro–financial risk and global volatility measures, such as sentiment only specifications, might be considered incomplete for crash prediction.

## *2.2. Crash Identification in Previous Studies*

Alongside predictive indicators, an essential aspect of the literature concerns the identification and dating of cryptocurrency crashes. Several studies have proposed model based approaches for detecting bubble bursts and abrupt market transitions. Shu et al. (2021) apply the Log-Periodic Power Law Singularity (LPPLS) model to detect bubble phases and subsequent crashes in Bitcoin, identifying multiple turning points during major boom–bust cycles. Similarly, Vladimir et al. (2022) combine log-periodic dynamics with AR CGARCH structures to forecast the December 2017 Bitcoin crash, illustrating the usefulness of tail-risk modelling for extreme event anticipation. In a related direction, Ismail et al. (2020) use topological data analysis to detect early warning signals before major drawdowns, highlighting the structural transitions in price networks that foreshadow crash events.

Empirical studies also document the timing and characteristics of major cryptocurrency crashes. Research consistently identifies pronounced crash episodes during the bursting of the 2017, 2018 bubble, the mid-2021 crash driven by China's regulatory crackdown, and the 2022 Terra-LUNA collapse, all of which feature abrupt price declines and volatility spikes (Conlon et al. 2020; Shaen et al. 2020). These episodes provide an important benchmark for evaluating crash identification strategies and emphasizes that cryptocurrency crashes typically unfold through a combination of speculative unwinding, deteriorating sentiment, and global risk amplification.

The literature reveals three recurring themes. cryptocurrency crashes are driven by the interplay of behavioral attention cycles, volatility spillovers, and macro-financial risk. Sentiment measures alone have limited predictive power unless combined with market based indicators. And the identification of crash events varies substantially between studies. These gaps motivate our structured crash identification framework and our systematic comparison of hybrid predictor sets using static and dynamic probit models.

### *2.3. Hypotheses Development*

The predictive modeling framework in this study is grounded in the premise that Bitcoin crashes arise from the joint interaction of behavioral sentiment, global market stress, and cryptocurrency specific dynamics. While sentiment driven measures such as Google search intensity capture shifts in investor attention, the empirical literature and our preliminary results indicate that sentiment alone yields weak and noisy crash signals. This is reflected in the inferior in-sample and out-of-sample performance of the sentiment-only specification (Model 1), which produces low AUC values and unstable probability paths.

In contrast, models incorporating macro-financial indicators, particularly the S&P 500 (SPX) and the Volatility Index (VIX) exhibit, substantially stronger predictive power. These variables capture spillovers from global equity markets and shifts in overall risk appetite, both of which have been shown to influence cryptocurrency downturns. Bitcoin returns, reflecting internal market dynamics such as deleveraging and momentum breakdowns, further enhance classification accuracy. The strongest performance across all specifications is achieved by the hybrid models (Model 4 and

Model 5), which combine sentiment, macro-financial variables, and Bitcoin specific information. These models consistently deliver the highest pseudo- $R^2$  and AUC values, with dynamic versions achieving AUC scores above 0.88 at medium-term horizons, confirming the multidimensional nature of crash formation. Motivated by these considerations, the following hypotheses are evaluated:

$H_0$  : Bitcoin crashes are not predictable using the available macro-financial, sentiment, or cryptocurrency specific indicators; none of the predictors have a statistically significant effect on crash probabilities.

$H_1$  : macro-financial indicators (SPX and VIX) significantly improve Bitcoin crash predictability by capturing global risk spillovers and market stress conditions, leading to superior pseudo- $R^2$  and AUC performance relative to sentiment only or price only models.

$H_2$  : Hybrid specifications that combine sentiment (GOOGLE), macro-financial indicators (SPX, VIX), and Bitcoin returns provide significantly higher predictive accuracy than models excluding one or more of these components, demonstrating that crash risk emerges from the interaction of behavioral attention, global uncertainty, and cryptocurrency specific dynamics.

### **3. Data and Methodology**

#### *3.1. Data*

Daily data were collected from January 1, 2017 till December 31, 2023, with 2556 observations from Bloomberg and the Federal Reserve Economic data. The total sample is divided into an in-sample period for estimation and an out-of-sample period for testing the model's forecasting ability. Our in-sample period is between 2017 and 2020, and the out-of-sample period is from 2021 until 2023. The study tests the following model combinations: Model 1: GOOGLE predictor only, Model 2: predictors SPX, GOOGLE, and VIX. Model 3: BTC and SPX predictors only, Model 4: All predictors with BTC, Model 5: Common factors variables with all predictors and BTC.

Google trends in Model 1 represent investor sentiment and public interest. The rationale behind using only Google trends in this model is to test the predictive power of sentiment data alone in

forecasting crashes.

Model 2 includes the S&P500 (SPX) and the Volatility Index (VIX) alongside Google trends. The rationale is to capture both broader market dynamics (SPX) and market volatility (VIX), which are known to influence risky assets like Bitcoin. VIX is often used as a measure of market fear and uncertainty, while SPX provides insight into overall market trends. Combining these predictors with Google trends aim to assess how sentiment and traditional market indicators contribute to crash prediction.

Model 3 uses Bitcoin price returns (BTC) and SPX. Here the aim is to directly analyze the relationship between Bitcoin's price movements and the general stock market. The stock market might influence Bitcoin's behavior due to broader economic factors, and this model evaluates whether Bitcoin's price trends in relation to SPX alone are sufficient for forecasting crashes.

Model 4 integrates all major predictors (SPX, GOOGLE, VIX, BTC) to provide a comprehensive view of the factors that could influence Bitcoin crashes. Combining sentiment (GOOGLE), market volatility (VIX), general market trends (SPX), and Bitcoin's price data (BTC), allows this model to find a complete picture of market conditions. The idea is that combining these factors can drive Bitcoin crashes, and develop accurate predictions.

Finally, Model 5 contains all the predictors with BTC and the factor models to capture broader economic, financial, and technological trends that could affect Bitcoin. These factors are derived from 43 variables grouped into market volatility, economic uncertainty, stock market indices, and commodity prices. The model tests the power to predict cryptocurrency crashes using macroeconomic and financial factors. Based on the methodology proposed by Bai et al. (2002) for determining the number of factors in a dataset, we applied their techniques to our dataset comprising 43 variables and 2556 daily observations. Upon application, it resulted in 7 significant factors. Factor 1: Market Volatility Indicators, Factor 2: Economic Uncertainty Measures, Factor 3: Monetary Policy and Economic Indicators, Factor 4: Stock Market Indices, Factor 5: Commodity Prices, Factor 6: Exchange rates, Factor 7: Digital Markets. See Table 9.

Table 1: Dataset

| Variables | Main Predictors                                                                                                                                                               |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Crashes   | The binary crash variable is constructed using the $3.09\sigma$ tail-event rule explained in Section 3.1. The crash series is aligned with all predictors at daily frequency. |
| BTC       | Bitcoin daily log returns                                                                                                                                                     |
| GOOGLE    | Google trends data for search terms "Bitcoin" and "cryptocurrency". Retrieved through API on R.                                                                               |
| SPX       | S&P500 daily log returns                                                                                                                                                      |
| VIX       | Global Volatility Index                                                                                                                                                       |

### 3.2. Methods

#### 3.2.1. Crash identification

Building on the literature, our approach develops a unified and robust crash identification strategy that avoids arbitrary thresholds and ensures comparability across horizons.

Our baseline crash definition employs a statistically motivated extreme value threshold similar to the tail-risk framework proposed in G et al. (2023). For each day  $t$ , we calculate the minimum one day return over the next  $h$  days,  $r_{\min}(t + 1, \dots, t + h)$ , and classify day  $t$  as a crash onset if:

$$r_{\min}(t + 1, \dots, t + h) \leq \mu_{\text{in}} - 3.09 \sigma_{\text{in}},$$

where  $\mu_{\text{in}}$  and  $\sigma_{\text{in}}$  are the mean and standard deviation of daily Bitcoin returns computed over the in-sample period, 2017 to 2020. The 3.09 coefficient corresponds to the 0.1% lower-tail of a normal distribution, producing a threshold that captures statistically extreme downside events consistent with the empirical behavior of prior cryptocurrency crashes documented in the literature.

To prevent double counting closely clustered tail events, each detected crash is expanded into a symmetric consolidation window of  $\pm w$  days. Any additional triggers within this window are suppressed, ensuring that each crash represents a distinct economic episode. This step reflects the evidence from previous bubble and crash studies that Bitcoin drawdowns often unfold over multiple days rather than occurring as isolated one day collapses.

### 3.2.2. Horizon and window grid search

Crash definitions can be sensitive to the choice of horizon  $h$  and consolidation window  $w$ . To avoid arbitrary specification choices, we perform an extensive grid search across:

$$h \in \{2, 5, 10, 20, 30\}, \quad w \in \{10, 20, 30\}.$$

For each combination, we construct the crash indicator and re-estimate all probit and dynamic probit models. Two patterns emerge. First, very short horizons, 2 to 5 days, generate extremely sparse crash labels, yielding negative pseudo- $R^2$  values and AUC scores close to 0.5, indicating effectively no discriminating ability. Second, horizons of 10, 20, and especially 30 days produce the strongest and most stable pseudo- $R^2$  and AUC values. This aligns with literature noting that cryptocurrency crashes typically unfold over multi-day windows linked to liquidity spirals, deleveraging, and sentiment deterioration (Vladimir et al. 2022; Shu et al. 2021).

Based on these results, the 30-day horizon with a  $\pm 30$ -day consolidation window emerges as the most reliable crash definition specification.

### 3.2.3. Validation against documented crash episodes

After identifying crash periods, we validate them against widely recognized real-world crash dates. We compare our model generated episodes to those reported in prior studies, such as the May 2021 China driven collapse, the 2022 Terra LUNA failure, and find strong alignment with documented market stress periods (Conlon et al. 2020; Shaen et al. 2020). We further cross-check our detected crash windows with crashes reported on CoinMarketCap,<sup>1</sup> a widely used source in cryptocurrency research, and with professional financial information platforms such as Bloomberg and Reuters.

Across all specifications, the crashes identified by our preferred rule correspond closely to the major downturns described in the academic literature and widely reported by industry sources. This strong concordance validates the economic relevance of the identified crash periods.

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<sup>1</sup>See <https://coinmarketcap.com/>

Figure 1 illustrates the resulting crash periods.

Table 2: Alignment of Model-Identified Crash Dates with Documented Market Crashes in Literature and financial resources

| Date         | Event                                                                                                                                                                                                             |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 26 Jan 2018  | the hack of Coincheck (Japan) caused sharp losses in Bitcoin and other digital-assets.                                                                                                                            |
| 11 June 2018 | Bitcoin falls after Coinrail hack documenting the major sell-off on 10–11 June 2018.                                                                                                                              |
| 27 Nov 2018  | Bitcoin declined from 6,500 USD to below 3,500 USD.                                                                                                                                                               |
| 8 July 2019  | Bitcoin surged to \$13,800 in late June 2019 before entering a multi-month correction. By July, August 2019, prices reversed sharply, initiating a significant drawdown replicated across major cryptocurrencies. |
| 12 Mar 2020  | Bitcoin plunges 40% in one of its biggest-ever daily crashes amid COVID-19 panic.                                                                                                                                 |
| 20 Dec 2020  | Major cryptocurrencies prices experienced a sharp decline.                                                                                                                                                        |
| 19 May 2021  | Bitcoin plunges 30% amid China crackdown and widespread liquidations.                                                                                                                                             |
| 5 July 2021  | Bitcoin dropped to 30,000 USD. Contributing factors included tweets by Elon Musk, concerns over Bitcoin's environmental impact, and China's crackdown on cryptocurrency mining and trading.                       |
| 23 Jan 2022  | After reaching another peak of around 69,000 USD in November 2021, Bitcoin fell to around 35,000 USD by late January.                                                                                             |
| 3 June 2022  | Bitcoin's price fell sharply to below 18,000 USD after the collapse of major crypto entities like Terra (LUNA) and its associated stablecoin UST. <sup>2</sup>                                                    |
| 30 Dec 2022  | By the end of 2022, cryptocurrencies showed varying degrees of price changes and market cap adjustments.                                                                                                          |
| 6 June 2023  | Cryptocurrency prices fall sharply after SEC sues Binance and Coinbase.                                                                                                                                           |

<sup>2</sup>Lee et al. (2023) discussed the crash of LUNA in their paper. They confirmed that the Terra-LUNA crash had a significant impact on the the overall cryptocurrency market.

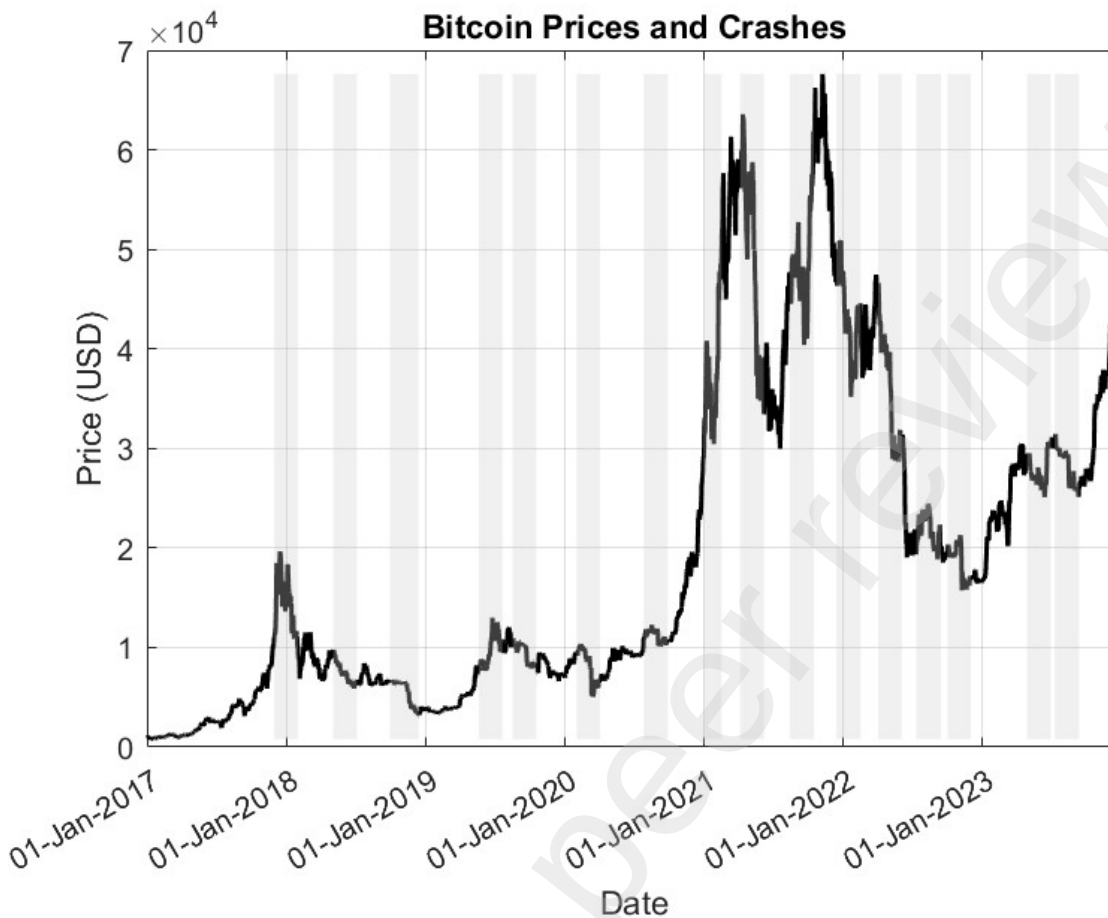


Figure 1: Crash episodes identified in Bitcoin returns

Note: This figure plots the time series of Bitcoin returns against crashes defined in gray shading as per crash rule of  $3.09\sigma$ , a 30 day window, and a 30 day horizon

### 3.3. Standard probit model

The probit model is a useful tool for predicting Bitcoin crashes, because of its statistical stability and compatibility with the distinct characteristics of the cryptocurrency market. Binary models such as probit can accurately represent the discrete nature of crash occurrences. They are essential in this highly unpredictable environment created by market dynamics. The model can directly assess the probability of a crash since it can handle binary outcomes, such as a crash or no crash. This is crucial in a market that frequently has abrupt and severe downturns that can happen nearly at any time. As a result, this model works well for binary outcomes and is suitable for estimating the probability of market crashes.

Consider the binary variable  $y_t$ :

$$y_t = \begin{cases} 1, & \text{if Bitcoin is in a crash at time } t; \\ 0, & \text{if Bitcoin has no crash at time } t \end{cases} \quad (1)$$

To do so, we consider a standard probit model in which the conditional probability of a crash event  $\{y_t = 1\}$  satisfies

$$\mathbb{E}_{t-1}[y_t] = \mathbb{P}_{t-1}(y_t = 1) = p_t = \Phi(\alpha + \delta BTC_{t-1} + z'_{t-1}\beta + f'_{t-1}\gamma), \quad (2)$$

where  $\Phi(\cdot)$  is the standard normal cumulative distribution function, ensuring that the conditional probability lies in the unit interval  $[0, 1]$ . The operators  $\mathbb{E}_{t-1}[\cdot]$  and  $\mathbb{P}_{t-1}(\cdot)$  denote, respectively, the conditional expectation and probability given information available at time  $t - 1$ . The variable  $BTC_{t-1}$  represents the lagged Bitcoin return,  $z_{t-1}$  is a vector collecting the other main predictors (such as SPX, VIX, and GOOGLE), and  $f_{t-1}$  is a vector of common factors extracted from a large panel of macro-financial variables.

This model includes past information on Bitcoin prices, and economic factors to estimate the likelihood of a crash at time  $t$ . Each coefficient changes the impact of these variables on our crash probability, providing insights into which factors are most predictive of market crashes.

To select the best model, we begin by choosing the one that provides the best in-sample forecast performance. Next, we assess the out-of-sample performance of the different models.

### 3.4. Performance measures

Model performance is evaluated using a combination of probability-forecast accuracy measures and classification metrics commonly employed in financial crisis prediction. For probability forecast accuracy, we report the pseudo- $R^2$ ,<sup>3</sup> the latter comparing the probit model to a constant probability

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<sup>3</sup>The pseudo- $R^2$  in probit models provides a measure of the goodness of fit relative to a model with no predictors beyond the constant. Unlike the R-squared used in linear regression, pseudo- $R^2$  values do not have a direct interpretation as a proportion of variance explained. Instead, they compare the likelihood function evaluated at the parameter estimates

benchmark. Higher pseudo- $R^2$  values indicate better calibration of predicted crash probabilities.

Since the primary objective of this study is to discriminate between crash and non-crash periods, we assess classification performance through the Receiver Operating Characteristic (ROC) curve and its Area Under the Curve (AUC). The ROC curve summarizes the trade-off between true-positive and false positive rates across all possible probability thresholds, while the AUC provides a threshold free measure of discriminatory ability. An AUC of 0.5 corresponds to random guessing, whereas values approaching 1 indicate strong predictive performance. The AUC metric is widely used in recession and financial crisis prediction studies (Nyberg 2010; Kauppi et al. 2008) and is particularly well suited to evaluating rare event forecasts such as cryptocurrency crashes.

Taken together, the pseudo- $R^2$  and AUC statistics provide complementary evidence on model performance, capturing both the calibration of predicted probabilities and the ability to discriminate between crash and non-crash periods. Throughout the analysis, the standard probit model serves as a benchmark against which the predictive gains of the dynamic probit specifications are assessed.

### *3.5. Robustness: Crash Definitions and Horizon Sensitivity*

To ensure that our crash labels are not sensitive to arbitrary modelling choices, we conduct a comprehensive set of robustness checks covering both alternative crash-definition rules and a full grid search over horizons and consolidation windows. We first consider three extensions to the baseline  $3.09\sigma$  tail-event rule. first, under a rolling-volatility approach,  $\mu$  and  $\sigma$  are computed using a 252-day window to allow thresholds to adapt to volatility-regime shifts, including the elevated uncertainty during the COVID-19 period. Then a cumulative drawdown rule flags a crash when the minimum cumulative return over the next  $h$  days falls below  $\mu_h - k\sigma_h$  or an absolute downturn (e.g.,  $-15\%$ ), capturing prolonged sell-offs not driven by a single extreme return. And finally a hybrid rule flags a crash if either the single-day  $3.09\sigma$  threshold or the cumulative-drawdown threshold is met. Across these definitions, the  $3.09\sigma$  rule combined with a 30-day horizon and a  $\pm 30$ -day consolidation window yields the most stable event frequency and strongest predictive performance

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to the likelihood of a restricted model with only an intercept. Typically, a higher pseudo- $R^2$  value indicates a better model fit to the observed data.

(pseudo- $R^2$  and AUC).

We systematically re-estimate all crash labels across horizons

$$h \in \{2, 5, 10, 20, 30\} \quad \text{and} \quad w \in \{10, 20, 30\},$$

recomputing the crash indicator for each  $(h, w)$  configuration. Very short horizons, 2 to 5 days, generate extremely weak signals, with pseudo- $R^2$  near zero or negative and AUC values close to 0.5. Predictive content emerges only at intermediate horizons, with  $h = 10, 20$ , and especially 30 days producing the highest and most stable discriminatory power across all probit specifications.

We further evaluate the sensitivity of results by comparing out-of-sample performance at horizons of 6, 9, 12, and 18 months. As in the crash-definition robustness, pseudo- $R^2$  values remain negative at very short horizons and become positive and economically meaningful only from six months onward. AUC values peak between six and twelve months, the range in which the hybrid predictor models (Models 4 and 5) consistently dominate.

Taken together, these robustness checks confirm that, the  $3.09\sigma$  tail-event rule, a 30-day crash horizon, and a  $\pm 30$ -day consolidation window constitute the most reliable and statistically coherent crash-definition framework for Bitcoin. This specification produces stable event frequencies, avoids label contamination, and provides the strongest in-sample and out-of-sample predictive performance across all model specifications.

#### 4. Empirical results

Table 3: Models Combination

| Model   | Predictors                                   |
|---------|----------------------------------------------|
| Model 1 | GOOGLE                                       |
| Model 2 | SPX GOOGLE VIX                               |
| Model 3 | BTC SPX                                      |
| Model 4 | SPX GOOGLE VIX BTC                           |
| Model 5 | Factor variables with all predictors and BTC |

#### 4.1. In-sample results

The in-sample performance of the five probit specifications is evaluated under the preferred crash definition ( $3.09\sigma$ , 30-day horizon, 30-day window). Table 4 shows that Models 2, 4, and 5 provide the strongest in-sample fit, with AUC values of 0.772, and 0.760 for models 4 and 5, alongside pseudo- $R^2$  values between 0.056 and 0.070. By contrast, the sentiment-only model (Model 1) and the BTC–SPX model (Model 3) deliver substantially weaker performance, with AUC values of 0.696 and 0.654.

Figure 2 illustrates the implied crash probabilities. Model 1 exhibits frequent and noisy probability spikes, reflecting the volatility of sentiment-driven signals. Model 3 produces smoother probability paths but systematically fails to capture major crash episodes. Models 4 and 5, which combine sentiment, macro–volatility indicators, and Bitcoin returns, generate the most coherent crash-probability trajectories with fewer false alarms and closer alignment with observed crash windows.

ROC Curves in 4 confirm this ranking. Models 2, 4, and 5 dominate the ROC space, while Models 1 remain below them throughout most thresholds. Among all specifications the in-sample results indicate that hybrid models combining sentiment and macro–financial risk indicators provide the most reliable explanatory power for Bitcoin crash risk.

| Model   | Pseudo- $R^2$ | AUC   |
|---------|---------------|-------|
| Model 1 | 0.032         | 0.696 |
| Model 2 | 0.056         | 0.772 |
| Model 3 | 0.025         | 0.654 |
| Model 4 | 0.070         | 0.760 |
| Model 5 | 0.069         | 0.760 |

Table 4: In-sample standard probit results for selected crash definitions.

#### 4.2. Out-of-sample results

Out-of-sample performance is examined across forecast horizons of 6, 9, 12, and 18 months. Consistent with the robustness analysis, pseudo- $R^2$  values remain negative for short horizons and turn positive only from six months onward (see Table 8 in appendix). Table 5 shows that Model 4 and

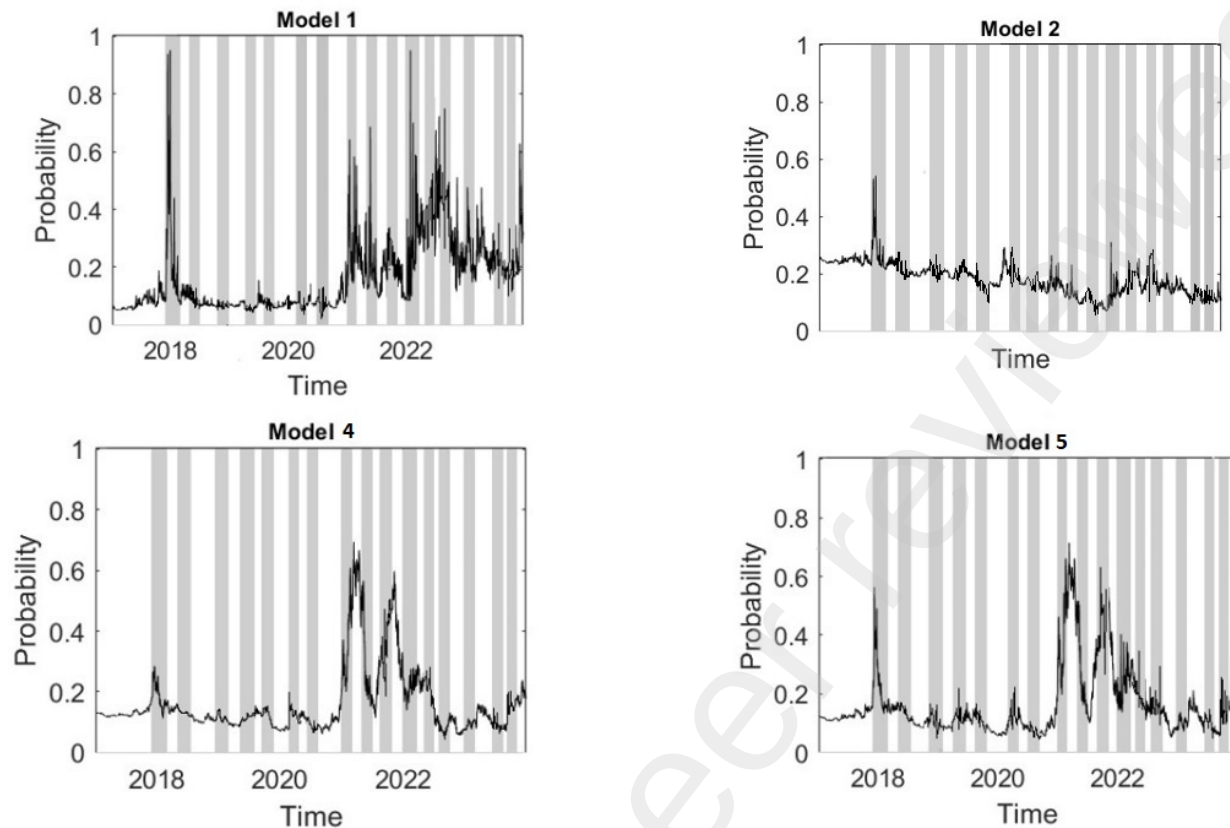


Figure 2: In-sample Model Fit

Note: This figure plots in-sample crash probabilities implied by selected models. Models 1, 2, 4, 5 are selected based on the value of AUC and pseudo- $R^2$  in table.

Model 5 deliver the strongest out-of-sample performance across all horizons, with almost similar AUC values of 0.831 and 0.832, and pseudo- $R^2$  values as high as 0.079 at the 12-month horizon.

In Table 8 Model 1 and Model 2 show considerably weaker performance, particularly at horizons below one year, where their pseudo- $R^2$  values often fall near zero or negative. Model 3 performs moderately well at 9 and 12 months (AUC between 0.781 and 0.812), but its pseudo- $R^2$  values remain smaller than those of Models 4 and 5.

Figure 3 shows that out-of-sample probability spikes are most coherent and correctly aligned with crash periods for Models 4 and 5. These models capture major turning points in 2021–2022 more effectively than factor free alternatives. The superior AUC of Model 4 (0.829 at 12 months) demonstrates that combining sentiment, global risk indicators (SPX, VIX), and Bitcoin returns generates the most consistent classification ability. Model 1, relying solely on Google Trends,

produces frequent but less accurate peaks, confirming that sentiment alone is insufficient for robust crash prediction.

Overall, the out-of-sample results indicate that the best performing specifications are the hybrid models with macro-financial and sentiment components, especially at medium term forecast horizons of six to twelve months.

| Model   | AUC   | Pseudo- $R^2$ |
|---------|-------|---------------|
| Model 1 | 0.830 | 0.028         |
| Model 2 | 0.828 | 0.017         |
| Model 3 | 0.826 | 0.015         |
| Model 4 | 0.832 | 0.032         |
| Model 5 | 0.831 | 0.031         |

Table 5: Out-of-sample standard probit AUC results for selected crash definitions.

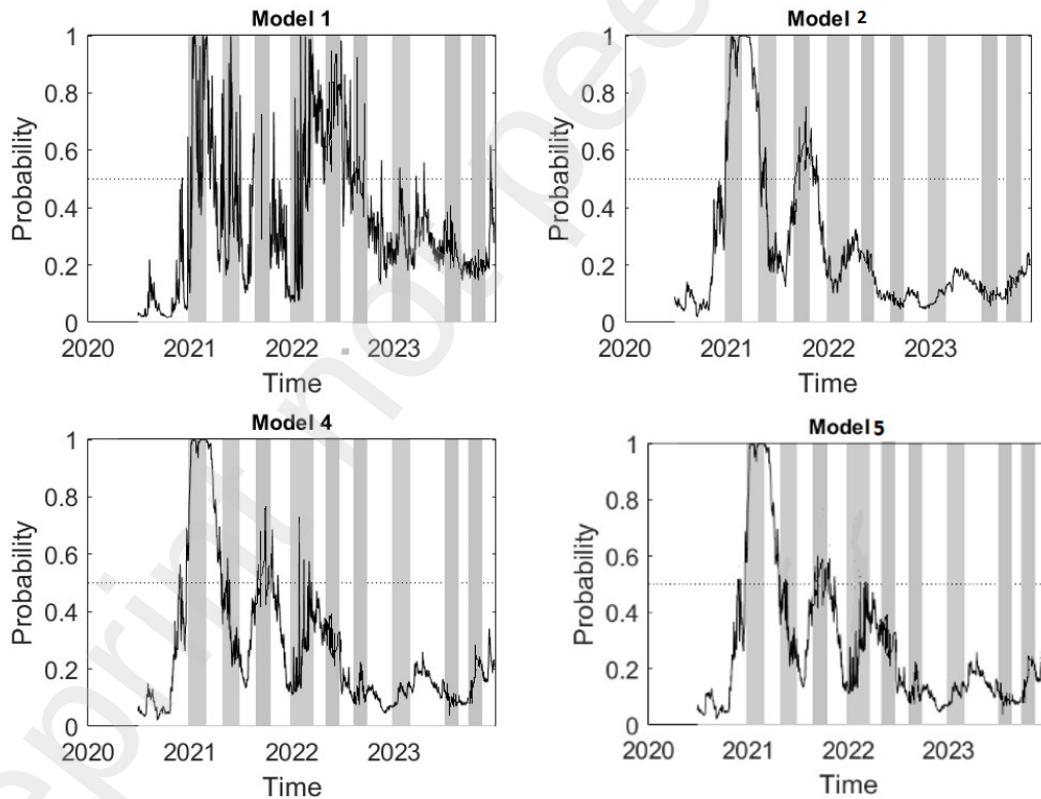


Figure 3: Out-of-sample Model Fit

Note: This figure plots out-of-sample crash probabilities implied by selected models. Models 1, 2, 4, 5 are selected are selected based on the value of AUC and pseudo- $R^2$ .

#### 4.3. Dynamic probit results

Dynamic probit models substantially improve predictive accuracy relative to their static counterparts. Table 6 reports that, in-sample, Models 2, 4, and 5 achieve the highest dynamic pseudo- $R^2$  and AUC values, outperforming Models 1 and 3. Dynamic Model 4 and 5 are the strongest performers in-sample, with an AUC of 0.772.

Out-of-sample performance Table 7 shows an even clearer improvement. Dynamic Models 4 and 5 achieve AUC values of 0.888 and 0.887, the highest among all specifications tested. These models also deliver consistently positive pseudo- $R^2$  values, while dynamic Models 1 and 3 perform notably worse. The improvement in AUC from 0.772 to 0.888 for Model 4 and 5 highlights the importance of including lagged latent dependence when modeling Bitcoin crash risk.

ROC curves in Figure 4 confirm that dynamic Models 4 and 5 dominate the static versions, producing higher true-positive rates across nearly all thresholds. This is consistent with the crash-persistence patterns identified in the robustness analysis: although lagged crash indicators ( $y_{t-1}$ ) have limited predictive power on their own, allowing the latent probit index to evolve dynamically captures time-varying crash risk more effectively.

In summary, dynamic probit models outperform standard probit models both in-sample and out-of-sample, with Models 4 and 5 providing the strongest and most stable predictive performance.

| Model   | Pseudo- $R^2$ | AUC   |
|---------|---------------|-------|
| Model 4 | 0.107467645   | 0.772 |
| Model 5 | 0.107467645   | 0.772 |
| Model 2 | 0.104393365   | 0.760 |

Table 6: Dynamic probit choosing the best in-sample results

| Model   | Pseudo- $R^2$ | AUC   |
|---------|---------------|-------|
| Model 1 | 0.020         | 0.820 |
| Model 2 | 0.029         | 0.832 |
| Model 3 | 0.017         | 0.785 |
| Model 4 | 0.035         | 0.888 |
| Model 5 | 0.036         | 0.887 |

Table 7: Out-of-sample dynamic probit AUC results for selected crash definitions.

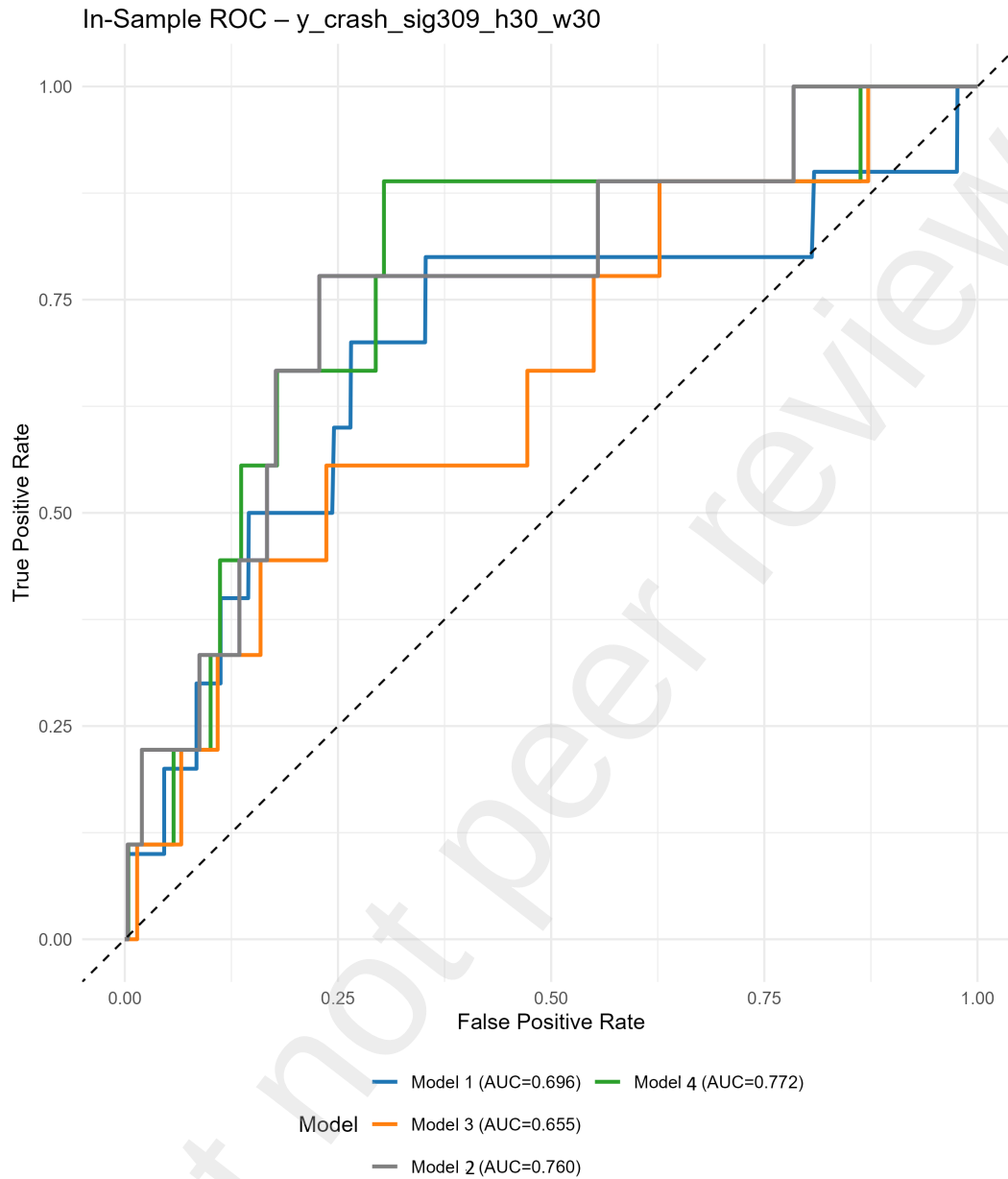


Figure 4: ROC best models of in-sample dynamic results

#### 4.4. Further Analysis

Across all empirical exercises, both in-sample and out-of-sample, and for both static and dynamic probit specifications, Models 4 and 5 consistently emerge as the strongest performers. These two hybrid models deliver the highest pseudo- $R^2$  values, the most coherent and well-calibrated crash-probability paths, and the top AUC scores across medium-term forecast horizons. Their

performance indicates that Bitcoin crash risk is best captured by a combination of behavioral attention, macro–financial stress, and cryptocurrency-specific dynamics, rather than by sentiment-only or narrow-market predictors.

Model 4 (SPX, GOOGLE, VIX, BTC) is the most parsimonious among the top performing models and demonstrates robust predictive power across all evaluation metrics. Its effectiveness stems from the integration of four complementary sources of crash risk. First, global financial stress, proxied by the S&P 500, captures cross-asset spillovers, reflecting the increasing integration of cryptocurrencies within broader risk-on/risk-off cycles. Second, the VIX index provides real-time information on global uncertainty and risk aversion factors that are highly correlated with speculative unwinding in cryptocurrency markets. Third, Google search intensity captures shifts in retail attention, sentiment, and fear, which are particularly influential in a market dominated by individual traders and momentum driven activity. Finally, Bitcoin returns incorporate endogenous market dynamics, such as leverage unwinding and liquidity driven sell-offs. Together, these channels generate a predictive structure capable of identifying crash conditions that emerge simultaneously from global stress, behavioral amplification, and internal fragility.

Model 5 extends this predictive framework by incorporating seven latent macro–financial factors extracted from a large panel of variables. Although Model 5 is more complex, it achieves comparable and in some cases slightly stronger out-of-sample AUC values, particularly in the dynamic specification, demonstrating that broader macro–financial information enhances classification accuracy. The strong performance of Model 5 reinforces the finding that cryptocurrency crashes are embedded within global financial conditions and are not isolated digital-asset phenomena.

The weaker performance of narrower models (Model 1 and Model 3) further supports this interpretation. Removing global risk factors (as in Model 1) or excluding sentiment (as in Model 3) reduces discriminatory power and increases noise in probability paths. This aligns with evidence from both the behavioral finance literature and macro–cryptocurrency spillover studies, which show that extreme cryptocurrency downturns typically arise from the interaction of global shocks, investor sentiment waves, and endogenous deleveraging.

Importantly, the superiority of Models 4 and 5 is also consistent with the selected crash definition framework. Under the  $3.09\sigma$  tail-risk rule with a 30-day horizon and  $\pm 30$ -day consolidation window, the resulting crash periods reflect episodes characterized by a combination of macro stress, sentiment deterioration, and momentum breakdown. The hybrid predictor sets in Models 4 and 5 are therefore naturally aligned with the empirical structure of these crashes, enhancing both calibration (pseudo- $R^2$ ) and classification accuracy (AUC).

Overall, the results indicate that Bitcoin crashes are multicausal events shaped simultaneously by global uncertainty, behavioral attention cycles, and internal market dynamics. Predictive modelling must therefore integrate heterogeneous predictors rather than rely on single theme approaches such as sentiment only or price only signals. As digital assets continue to integrate with traditional financial markets, hybrid frameworks like Models 4 and 5 will become increasingly relevant for risk management, regulatory monitoring, and early warning system design.

## 5. Conclusion

This study examined the predictability of Bitcoin crashes using standard and dynamic probit models and a range of behavioral, macro-financial, and cryptocurrency specific predictors. Our analysis contributes to the literature in three main ways. First, we develop a robustness based crash identification framework that systematically compares thresholds, horizons, and consolidation windows. By combining a  $3.09\sigma$  tail-risk rule with a comprehensive grid search, the proposed approach resolves inconsistencies in prior cryptocurrency crash definitions and generates economically meaningful and statistically robust event labels.

Second, we provide the first structured comparison of sentiment based, macro-financial, and cryptocurrency specific predictors within a unified probit framework. The results show that hybrid models, particularly those integrating Google search intensity, S&P 500, VIX, and Bitcoin returns, consistently outperform sentiment only or narrow predictor specifications. This indicates that Bitcoin crash risk is driven not by investor attention alone, but by the interaction of global market stress, volatility conditions, and endogenous cryptocurrency dynamics.

Third, the findings reveal that Bitcoin crash predictability is highly horizon dependent. Meaningful discriminatory power emerges only at intermediate horizons of 10 to 30 days, whereas very short horizons exhibit no predictive ability. This pattern underscores the importance of horizon selection when evaluating crash risk in high-volatility digital asset markets and reconciles conflicting results in the existing literature.

In conclusion, the analysis shows that cryptocurrency crashes can be anticipated when models combine behavioral indicators with macro-financial risk signals and when crash events are defined in a robust and economically grounded manner. These findings have implications for investors, exchanges, and regulators seeking forward looking measures of systemic risk in digital asset markets. Future research may extend this framework to include high frequency liquidity measures, on-chain indicators, or cross asset contagion channels, offering further insights into the dynamics of cryptocurrency market stress.

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data supporting the findings of this study are available from Bloomberg, FRED, Google API, and CoinMarketCap.

## **Appendix A   Forecasting Results**

| <b>Horizon</b>   | <b>Model</b> | <b>Pseudo-<math>R^2</math></b> | <b>AUC</b> |
|------------------|--------------|--------------------------------|------------|
| <b>6 months</b>  |              |                                |            |
|                  | Model 4      | 0.062                          | 0.808      |
|                  | Model 5      | 0.062                          | 0.808      |
|                  | Model 2      | 0.026                          | 0.732      |
|                  | Model 1      | 0.024                          | 0.693      |
|                  | Model 3      | -0.003                         | 0.760      |
| <b>9 months</b>  |              |                                |            |
|                  | Model 4      | 0.067                          | 0.805      |
|                  | Model 5      | 0.067                          | 0.805      |
|                  | Model 2      | 0.026                          | 0.652      |
|                  | Model 3      | 0.024                          | 0.781      |
|                  | Model 1      | 0.023                          | 0.646      |
| <b>12 months</b> |              |                                |            |
|                  | Model 4      | 0.079                          | 0.829      |
|                  | Model 5      | 0.079                          | 0.829      |
|                  | Model 3      | 0.047                          | 0.812      |
|                  | Model 1      | 0.029                          | 0.683      |
|                  | Model 2      | 0.027                          | 0.664      |
| <b>18 months</b> |              |                                |            |
|                  | Model 4      | 0.045                          | 0.743      |
|                  | Model 5      | 0.045                          | 0.743      |
|                  | Model 3      | 0.041                          | 0.804      |
|                  | Model 1      | -0.009                         | 0.560      |
|                  | Model 2      | -0.012                         | 0.538      |

Table 8: Out-of-sample pseudo- $R^2$  and AUC across horizons and models.

## Appendix B Factor Variables

| Factor 1 Market Volatility Indicators                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CBOE gold ETF Volatility Index<br>CBOE EuroCurrency ETF Volatility Index<br>CBOE Energy Sector ETF Volatility Index<br>CBOE Volatility Index<br>CBOE Silver ETF Volatility Index<br>CBOE Crude Oil ETF Volatility Index<br>Equity Market Volatility Infectious Disease Tracker<br>CBOE S&P 500 3-Month Volatility Index<br>ICE BofA Asia Emerging Markets Corporate Plus Index Effective Yield<br>CBOE Equity VIX on Google                              |
| Factor 2 Economic Uncertainty Measures                                                                                                                                                                                                                                                                                                                                                                                                                   |
| US Economic Policy Uncertainty Index<br>Equity Market-related Economic Uncertainty Index                                                                                                                                                                                                                                                                                                                                                                 |
| Factor 3 Monetary Policy and Economic Indicators                                                                                                                                                                                                                                                                                                                                                                                                         |
| Federal Funds Target Range - Upper Limit<br>ICE BofA BBB US Corporate Index Effective Yield<br>ICE BofA US Corporate Index Option-Adjusted Spread<br>15-Year Fixed Rate Conforming Mortgage Index<br>USD INDEX<br>Interest Rates<br>Federal Funds Target<br>ECB Deposit Facility Rate for Euro Area<br>ICE BOFA US High Yield Index Effective Yield (debt)<br>Market Yield on U.S. Treasury Securities at 10-Year Constant Maturity<br>Inflation-Indexed |
| Factor 4 Stock Market Indices                                                                                                                                                                                                                                                                                                                                                                                                                            |
| DowJones<br>Nikkei 225<br>NASDAQ<br>China Market Cap<br>Shanghai Composite Index<br>FTSE 100 Index                                                                                                                                                                                                                                                                                                                                                       |
| Factor 5 Commodity Prices                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Oil - West Texas I<br>GOLD<br>Brent Oil<br>Silver                                                                                                                                                                                                                                                                                                                                                                                                        |
| Factor 6 Exchange rates                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| BTCUSD<br>Yuan<br>Yen<br>Pounds<br>EUSD                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Factor 7 Digital Markets                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Bitcoin Futures CME<br>ECPI Global ESG Smart Cities<br>USDT (stablecoins)<br>Coinbase Bitcoin Cash<br>Coinbase Index<br>Negative Tweets on Bitcoin                                                                                                                                                                                                                                                                                                       |

Table 9: Economic and Financial Factor Indicators

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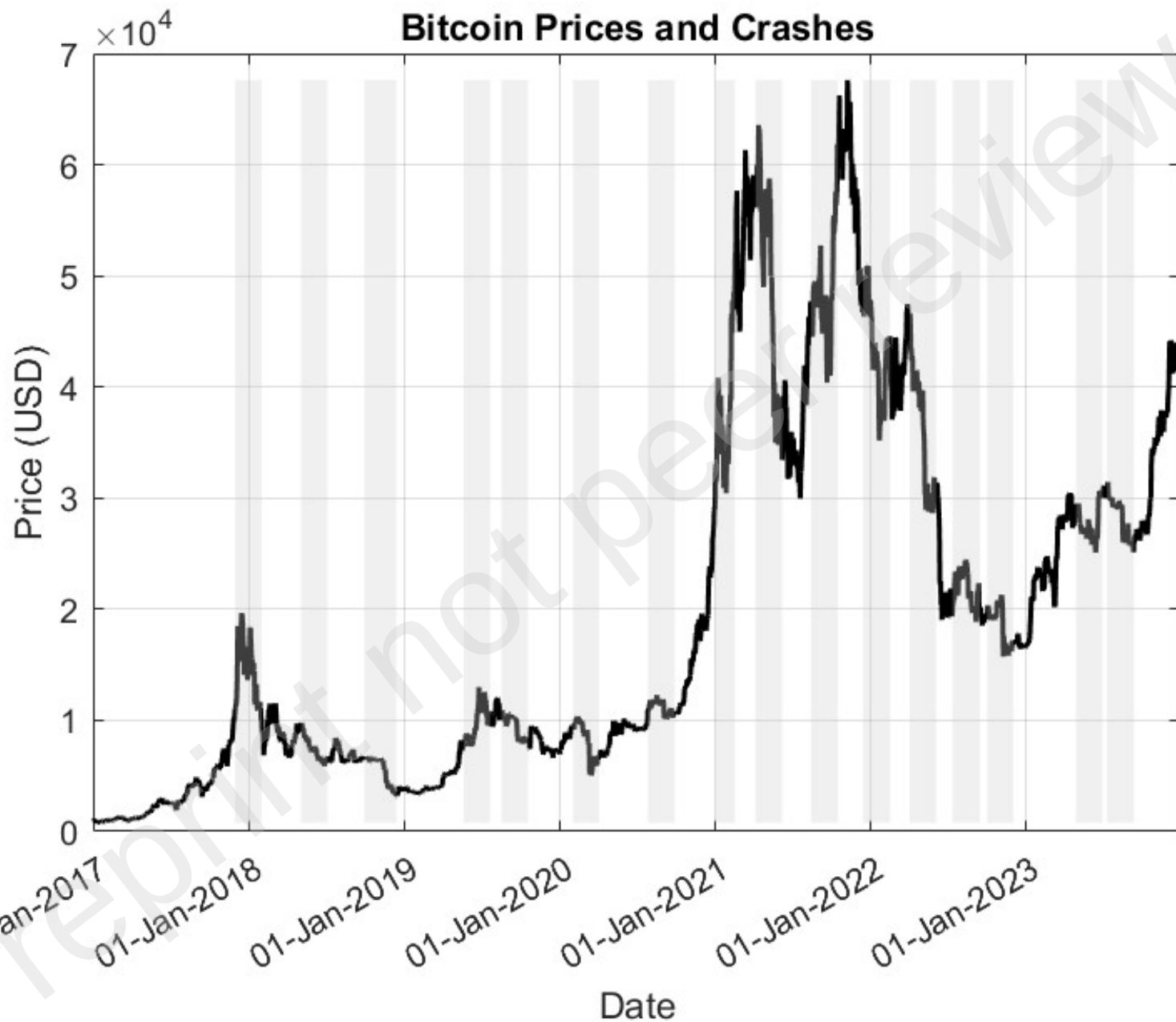
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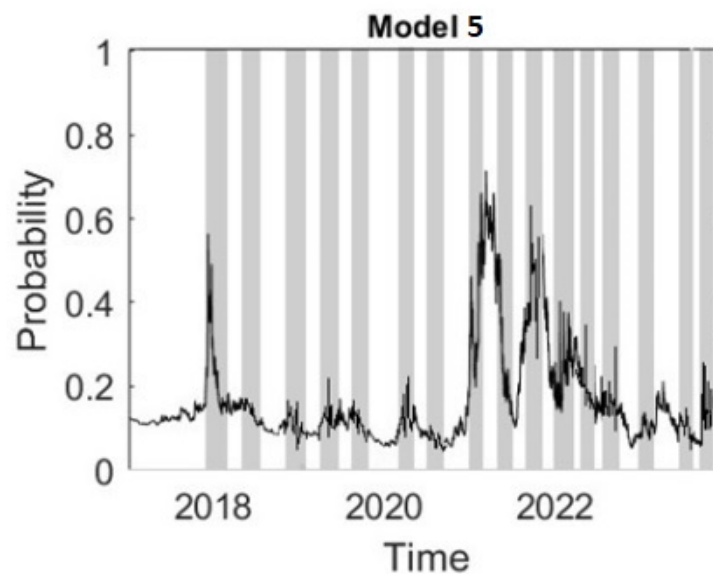
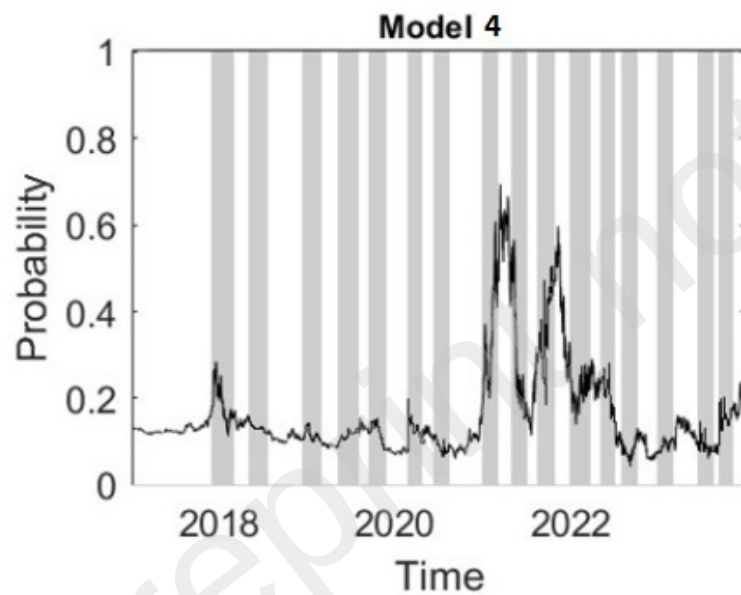
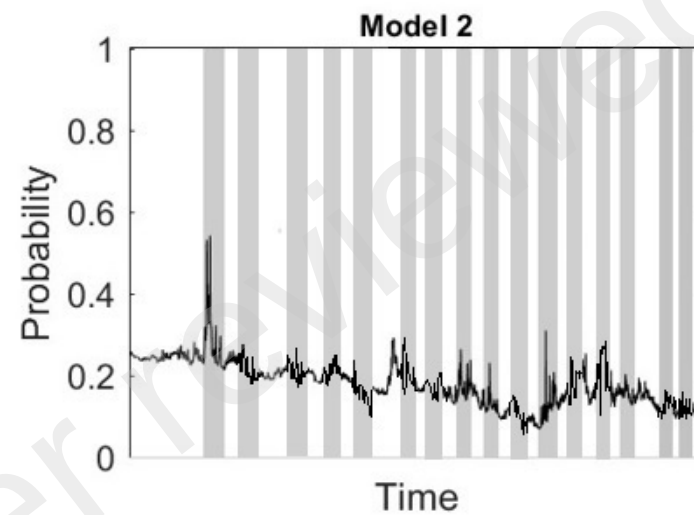
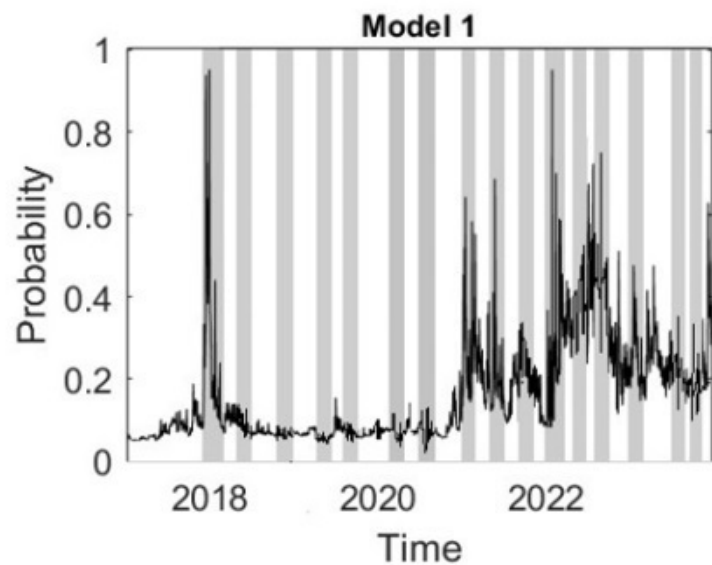
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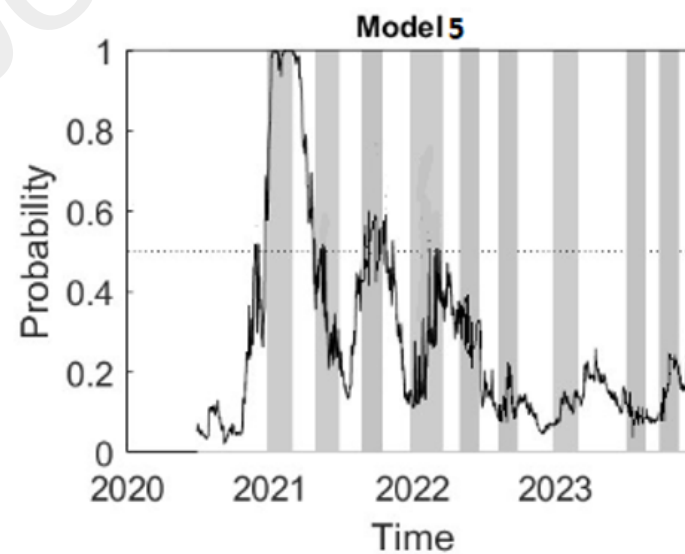
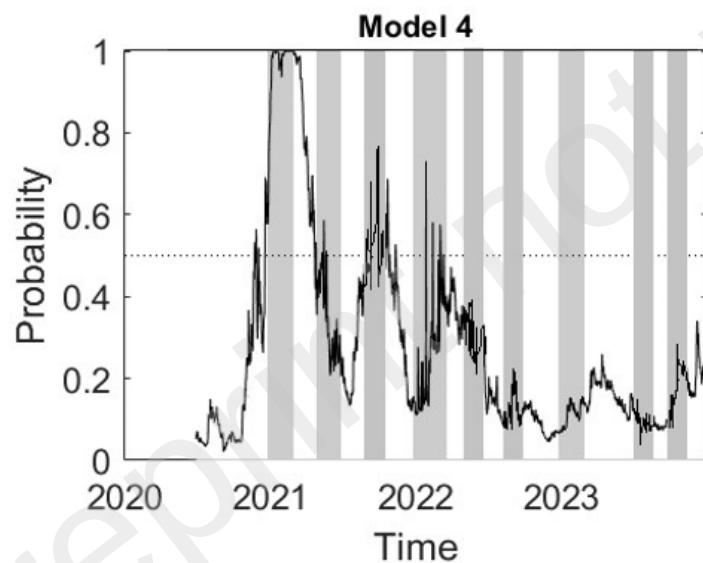
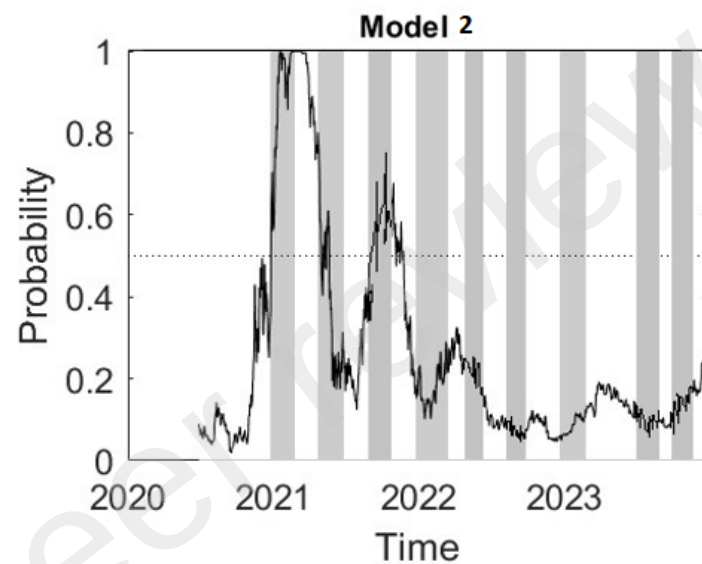
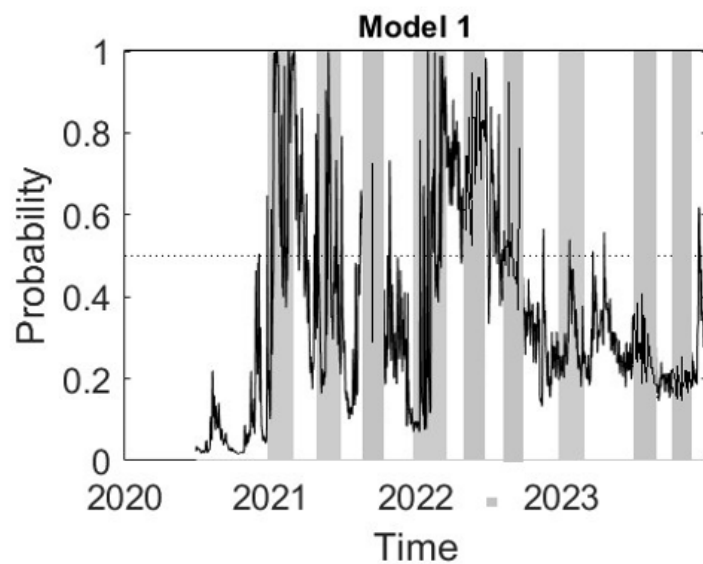
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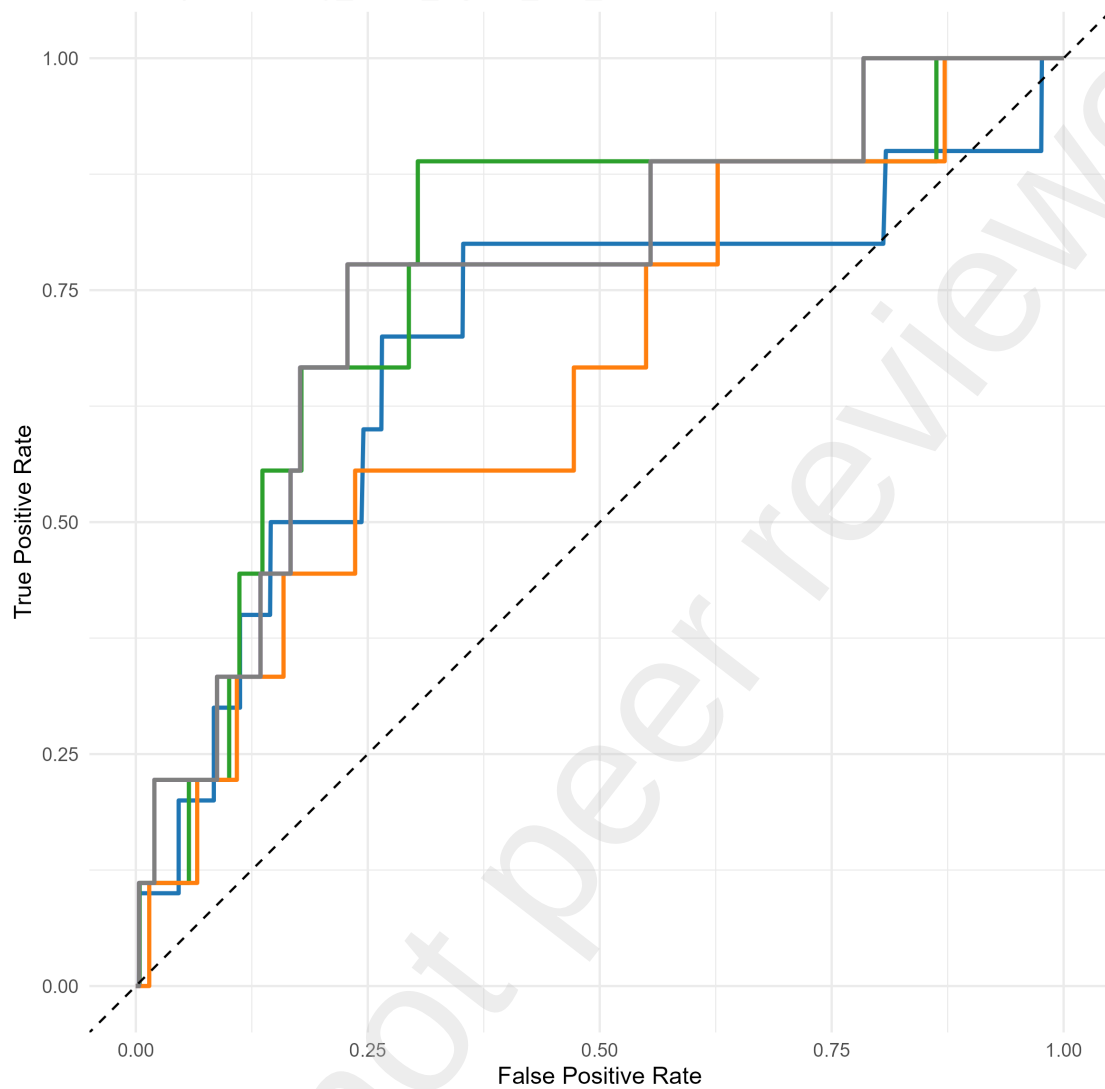
**Bitcoin Prices and Crashes**







In-Sample ROC – y\_crash\_sig309\_h30\_w30



Model 1 (AUC=0.696) Model 4 (AUC=0.772)  
Model 3 (AUC=0.655) Model 2 (AUC=0.760)